Mean accuracy: +1.44 points against v3

Fixed comparison: 33 labels, +1/+2/+5 seconds, 45 shared validation scenario groups

Horizon	Model	Top-1	Top-3	F1 / 33	NLL	Brier	ECE
+1 s	Retained v3	65.62%	90.10%	0.4120	1.1155	0.4694	2.94 pp
+1 s	Frozen v4	67.09%	91.01%	0.4125	1.0688	0.4515	4.16 pp
+2 s	Retained v3	65.17%	89.82%	0.4115	1.1324	0.4738	2.93 pp
+2 s	Frozen v4	66.62%	90.78%	0.4180	1.0855	0.4564	3.63 pp
+5 s	Retained v3	63.48%	89.25%	0.3665	1.1642	0.4916	3.42 pp
+5 s	Frozen v4	64.86%	89.86%	0.3767	1.1243	0.4763	4.36 pp

Paired scenario-bootstrap gains (95% intervals):
+1 s: +1.48 pp [+0.92, +2.11] | +2 s: +1.45 pp [+0.89, +2.09] | +5 s: +1.38 pp [+0.77, +2.11]

Mean gain 95% interval: [+0.90, +2.08] pp. Predeclared +1.00 pp target: met. Active budget: 15 models / 31.73M parameters; v3: 12 / 28.98M.

ECE worsened at all three horizons (2.94/2.93/3.42 → 4.16/3.63/4.36 pp at +1/+2/+5 s); NLL and Brier improved.

F1 averages all 33 labels. Validation has been inspected before: these are development findings, not final test results.

Eight preregistered candidates, one frozen decision

Screening and confirmation remain entirely within the original 210 training scenario groups

Candidate	Weight	Top-1	Δ pp	NLL	Guard
20 s multiscale MLP	0.25	59.21%	+0.12	1.299	yes
20 s TCN	0.25	59.29%	+0.20	1.266	yes
20 s GRU	0.25	59.30%	+0.21	1.268	yes
60 s multiscale MLP	0.25	59.39%	+0.30	1.296	yes
60 s TCN	0.50	60.06%	+0.96	1.260	yes
60 s GRU	0.25	59.41%	+0.32	1.248	yes
60 s MLP wide	0.50	59.29%	+0.20	1.315	yes
20 s MLP deep	0.25	59.42%	+0.33	1.297	yes

Screen uses a four-member seed-42 reference. Each row shows its best eligible registered blend (or best unguarded result if none passed). These are not validation scores.

Confirmed blend	w	Δ pp	Both
60 s TCN	0.50	+0.88	yes
60 s TCN	0.75	+0.61	yes
equal pair	0.75	+0.56	yes
60 s TCN	0.25	+0.55	yes
equal pair	1.00	+0.54	yes
equal pair	0.50	+0.54	yes

Frozen: 60 s TCN; new-member weight 0.50.

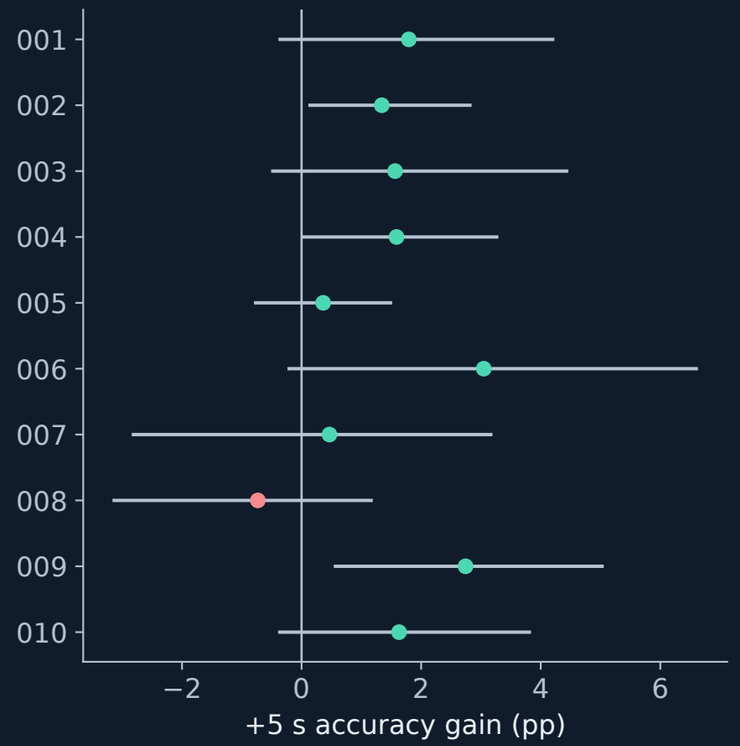
Confirmation uses all 12 internal v3 members. Each fold must show positive mean accuracy gain, NLL ≤ baseline +0.02, F1 ≥ baseline −0.01 and each horizon ≥ baseline −0.003.

147 fit / 21 stop / 42 select groups per internal partition. Normalization uses fit groups; epochs and temperature use stop groups. The same two partitions overlap and have informed earlier experiments.

+5 s improves for 9 of 10 pilots

Same ten trained policies; paired intervals keep shared scenario groups together

Pilot	Origins	v3 top-1	v4 top-1	Gain pp	95% interval
001	2,845	54.06%	55.85%	+1.79	[-0.39, +4.23]
002	3,057	75.60%	76.94%	+1.34	[+0.11, +2.84]
003	2,878	63.90%	65.46%	+1.56	[-0.51, +4.46]
004	2,897	67.79%	69.38%	+1.59	[+0.00, +3.29]
005	3,049	74.16%	74.52%	+0.36	[-0.80, +1.52]
006	2,987	62.34%	65.38%	+3.05	[-0.24, +6.63]
007	3,001	54.42%	54.88%	+0.47	[-2.84, +3.19]
008	2,871	46.08%	45.35%	-0.73	[-3.16, +1.19]
009	2,990	69.03%	71.77%	+2.74	[+0.54, +5.05]
010	2,883	66.08%	67.71%	+1.63	[-0.39, +3.84]



Pilot difficulty differs because behavior and label prevalence differ. These results do not test unseen pilots. Pointwise intervals are descriptive and are not corrected for examining ten pilots and three horizons.

Per-skill F1 preserves the full 33-label catalogue

+5 seconds; support counts eligible forecast origins, which are correlated within flights

Skill	n	v3 F1	v4 F1	Δ F1
abort_support	113	0.521	0.515	-0.007
accelerate_to_speed	0	—	—	—
beam_missile_left	6	0.250	0.000	-0.250
beam_missile_right	20	0.203	0.513	+0.309
beam_target_left	1,693	0.712	0.725	+0.013
beam_target_right	1,445	0.714	0.737	+0.022
climb_to_altitude	1	0.000	0.000	+0.000
commit	0	—	—	—
crank_target_left	1,150	0.648	0.653	+0.004
crank_target_right	947	0.790	0.788	-0.002
decelerate_to_speed	126	0.326	0.307	-0.019
defensive_reversal	1	0.000	0.000	+0.000
descend_to_altitude	162	0.619	0.631	+0.012
dive_defense	0	—	—	—
drag_missile	0	—	—	—
extend	2,676	0.592	0.602	+0.010
lag_pursuit	4,872	0.614	0.632	+0.018

Skill	n	v3 F1	v4 F1	Δ F1
last_ditch_break	0	—	—	—
launch	130	0.494	0.507	+0.013
lead_pursuit	1,350	0.539	0.543	+0.004
lock_target	0	—	—	—
maintain_heading	36	0.553	0.500	-0.053
notch_left	124	0.400	0.430	+0.030
notch_right	80	0.444	0.462	+0.017
preemptive_crank	0	—	—	—
pursue_target	2,561	0.565	0.562	-0.003
recommit	4,857	0.650	0.666	+0.016
search	2	0.000	0.000	+0.000
secondary_shot	1,414	0.647	0.656	+0.009
short_range_attack	216	0.305	0.291	-0.014
support_missile	19	0.320	0.429	+0.109
turn_cold	5,437	0.679	0.701	+0.023
turn_to_heading	20	0.510	0.585	+0.076

Dash = no true validation examples at this horizon; it is not evidence of perfect or failed recall. Catalogue macro-F1 retains all 33 labels, including absent labels. Rare-skill changes need support-aware interpretation.

Complete precision, recall, F1 and predicted counts for every pilot × horizon × skill are in the evidence CSVs; skill_support.csv adds flight, scenario-group and pilot coverage.

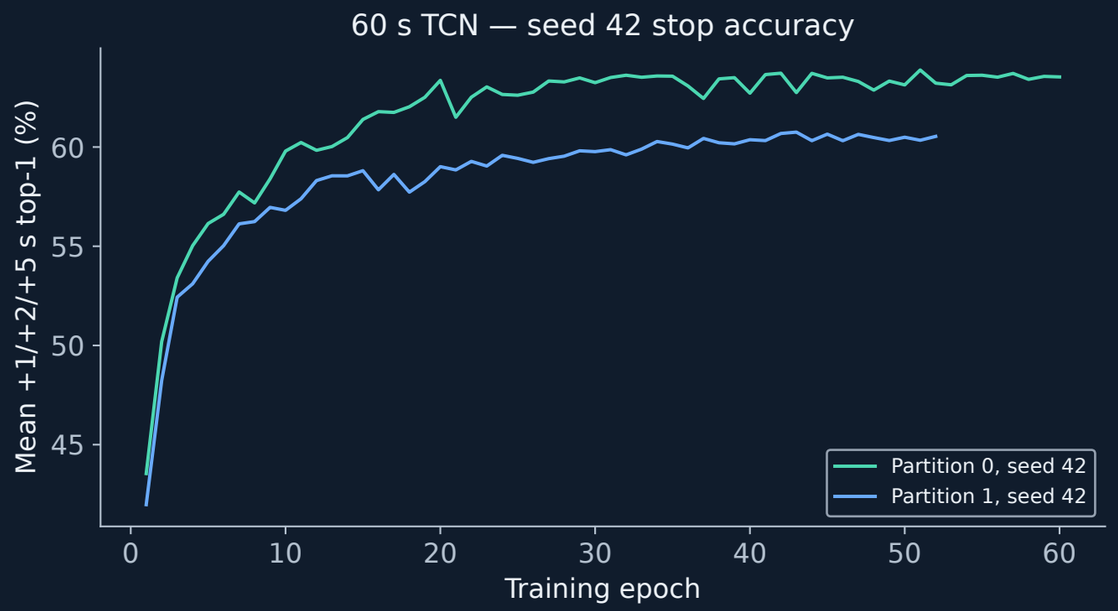
Training cost and the evidence needed to reproduce this run

Complete runs and unsuccessful candidates are retained; the previous slide pack and baseline remain unchanged

Phase	Runs	Logged time	Epochs
internal	34	49.50 min	1626
refit	3	7.59 min	126

37 completed new fits; 8 cached internal v3 runs. Total logged training time: 57.09 minutes. Timers exclude data loading, features and setup. Some confirmation fits shared the GPU concurrently: summed timers include contention, not end-to-end wall time or isolated-model throughput.

Selected budget: 15 models, 31.73M parameters. Final models use fixed epochs and temperatures chosen internally. Final refits have training-objective logs only; outer-validation curves are deliberately absent.



Shown: seed 42 in each partition, not an average over seeds. Stop curves choose epochs within training-only partitions. Complete objectives, NLL, seed variation and timing are exported in training_histories.csv and run_inventory.csv.

Verified: all 490 pre-existing artifact files unchanged; frozen sources and partitions match; validation origins align. Rebuild commands and source hashes accompany the PDF. No reserved test was evaluated.